

ELY, MN, USA

WMO: 727459

Lat: 47.817N

Lon: 91.833W

Elev: 444

StdP: 96.11

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94964

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.5	-28.8	-37.0	0.1	-32.6	-32.8	0.2	-28.3	8.5	-11.3	7.7	-11.1	0.6	250	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	29.0	20.4	27.6	19.4	26.2	18.3	22.9	27.0	21.3	25.6	20.0	24.1	4.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.9	25.5	19.8	15.3	23.7	18.6	14.2	22.7	69.9	26.8	63.8	25.2	59.4	24.0	35.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	7.1	6.0	-36.4	32.2	3.8	2.0	-39.1	33.6	-41.3	34.7	-43.5	35.9	-46.2	37.3		
(5)			-36.4	24.9	3.5	3.1	-38.9	27.1	-40.9	28.9	-42.9	30.6	-45.4	32.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-14.1	-11.7	-4.5	3.0	10.3	15.8	18.5	17.2	13.0	5.8	-2.4	-10.3
	DBStd	12.75	8.22	7.75	7.42	5.38	4.83	3.83	3.37	3.41	4.68	4.85	6.39	7.52
	HDD10.0	3254	748	608	451	217	54	3	0	1	22	151	372	629
	HDD18.3	5531	1006	841	708	459	253	92	41	63	170	389	621	888
	CDD10.0	870	0	0	2	8	64	178	263	224	111	20	1	0
	CDD18.3	106	0	0	0	0	5	18	45	28	10	1	0	0
	CDH23.3	1158	0	0	0	3	91	226	451	291	90	7	0	0
	CDH26.7	253	0	0	0	0	20	53	104	59	16	1	0	0
Wind	WSAvg	2.4	2.3	2.3	2.5	2.6	2.7	2.4	2.2	2.1	2.4	2.7	2.6	2.2
Precipitation	PrecAvg	736	22	18	29	57	77	102	102	82	101	74	41	28
	PrecMax	914	58	46	58	126	120	199	197	181	275	134	101	54
	PrecMin	590	8	6	9	15	31	19	43	23	20	30	8	6
	PrecStd	89	12	10	14	34	24	42	41	39	60	29	23	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	7.9	17.2	22.6	29.0	31.8	32.0	31.1	28.3	23.9	15.2	6.2
		MCWB	0.9	4.3	11.3	12.8	17.7	23.6	22.1	22.2	20.0	16.0	10.3	2.7
	2%	DB	1.4	4.8	12.2	18.7	26.1	27.8	29.0	27.9	26.1	19.9	12.2	3.0
		MCWB	0.2	2.4	7.1	10.3	15.4	18.8	21.3	20.1	18.7	13.7	9.1	1.3
	5%	DB	-0.9	2.3	8.7	16.2	22.8	26.1	27.5	26.4	23.7	17.2	8.0	2.0
		MCWB	-1.5	0.3	4.9	8.2	13.9	17.5	20.0	19.0	17.6	12.3	5.3	0.8
	10%	DB	-2.5	-0.1	6.0	12.8	20.1	23.9	26.1	24.8	21.7	13.8	6.3	0.0
		MCWB	-3.3	-1.3	2.4	6.2	12.7	16.5	18.8	18.3	16.5	9.7	3.8	-0.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	5.1	12.7	14.8	20.0	24.1	24.7	24.3	22.7	17.4	12.0	3.5
		MCDB	3.7	7.3	16.2	20.1	25.1	28.9	28.9	28.4	26.7	20.9	14.1	5.0
	2%	WB	0.2	2.5	7.9	10.8	17.3	21.2	23.1	21.8	20.3	14.9	8.6	1.9
		MCDB	1.3	4.6	10.7	16.9	22.9	26.2	27.0	26.0	23.9	18.7	11.5	3.3
	5%	WB	-1.5	0.4	4.9	8.9	15.3	19.5	21.5	20.4	18.8	12.6	6.0	0.4
		MCDB	-0.9	2.0	8.5	15.1	21.0	23.5	25.7	24.7	22.0	16.8	8.2	1.5
	10%	WB	-3.3	-1.6	2.7	6.9	13.6	17.9	20.1	19.3	17.3	10.4	4.0	-1.0
		MCDB	-2.4	-0.5	5.5	12.4	18.6	21.7	24.3	23.3	20.4	13.4	6.1	-0.2
Mean Daily Temperature Range	5% DB	MDBR	11.3	12.6	12.5	12.4	13.2	12.5	12.3	12.4	11.6	9.4	8.1	9.1
		MCDBR	10.2	12.4	14.1	17.9	17.9	15.4	13.7	14.4	14.5	14.5	11.3	10.1
		MCWBR	9.0	9.8	9.4	10.6	9.3	7.9	7.1	7.7	8.5	9.0	8.1	8.4
	5% WB	MCDBR	9.5	10.5	13.3	16.0	14.5	12.2	12.1	12.3	11.4	13.1	10.7	9.0
		MCWBR	8.4	8.3	9.5	10.2	8.2	7.3	7.3	7.5	7.5	8.8	8.0	7.6
Clear-Sky Solar Irradiance	taub	0.249	0.264	0.307	0.335	0.355	0.370	0.383	0.369	0.352	0.308	0.284	0.257	
	taud	2.381	2.345	2.375	2.423	2.418	2.394	2.383	2.451	2.474	2.582	2.528	2.428	
	Ebn at Noon	851	910	916	919	907	892	874	874	857	855	804	794	
	Edh at Noon	67	87	100	106	111	115	115	103	91	70	59	57	
All-Sky Solar Radiation	RadAvg	1.28	2.35	3.59	4.70	5.33	5.70	6.00	5.09	3.75	2.23	1.23	0.95	
	RadStd	0.16	0.25	0.33	0.47	0.36	0.43	0.28	0.36	0.27	0.28	0.15	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-53
(47) Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page